EXATHON INDIA 2026 · TRACK 01

Trace*Hold*

Trace the stolen money. Hold only what the records support.

THE WORKED EXAMPLE

	FROZEN	SUPPORTED
Account A	₹30,000	₹30,000
Account B	₹1,90,000	₹70,000
Account C	₹4,50,000	₹50,847
Total	**₹6,70,000**	**₹1,50,847**

₹5,19,153 of the account holders' own money would be held without support.

Lisa Mehta · Built on Exasol Personal · github.com/lisamehta0791/TraceHold

Holds come too late for victims, and too broad for everyone else

IN ONE SENTENCE

An officer deciding how much of an account to hold in a cyber-fraud case can't tell how much of the victim's stolen money actually reached that account, because the money is split, mixed with other funds and moved on within minutes, and requests often arrive without the amount or the trail.

If this worked, holds would match what the records support, the victim's money would be held sooner, and a person who received one payment would not have their whole balance locked.

₹1,261 cr

lost to online fraud in Bengaluru up to Aug 2025, only about 4% traced and returned

Too late for victims · Deccan Herald, city police data

105 petitions

whole accounts frozen for disputed sums sometimes under ₹100, freezes continuing after a clean chit

Too broad for innocent recipients · Rajasthan High Court, Aug 2026 (as reported)

Five people live with this problem every day

	WHAT THEY NEED	WHAT GOES WRONG TODAY
Fraud victim	Wants the money held before it moves on	Recovery is slow and low; the hold depends on how fast the trail is understood
Holder of a downstream account	A restriction limited to what is disputed	Whole-balance freezes, often without notice or a stated amount
Officer at the police cyber cell	To state the amount and basis of a hold	The trail spans many accounts; mixed funds blur the disputed share
Bank nodal officer	To act lawfully and promptly	No way to check a requested amount, so blanket freezing is the safe default
State oversight officer	To spot prolonged or disproportionate freezes	No consolidated view of how long holds last or how big they are

The rule exists. The calculator does not.

1

Money moves in minutes

Requests are made before the full trail is known.

2

Requests arrive incomplete

Missing reference, legal provision, transaction or amount.

3

Mixed money has no label

Which part is 'disputed' depends on a convention; without one, the only safe choice is the whole balance.

4

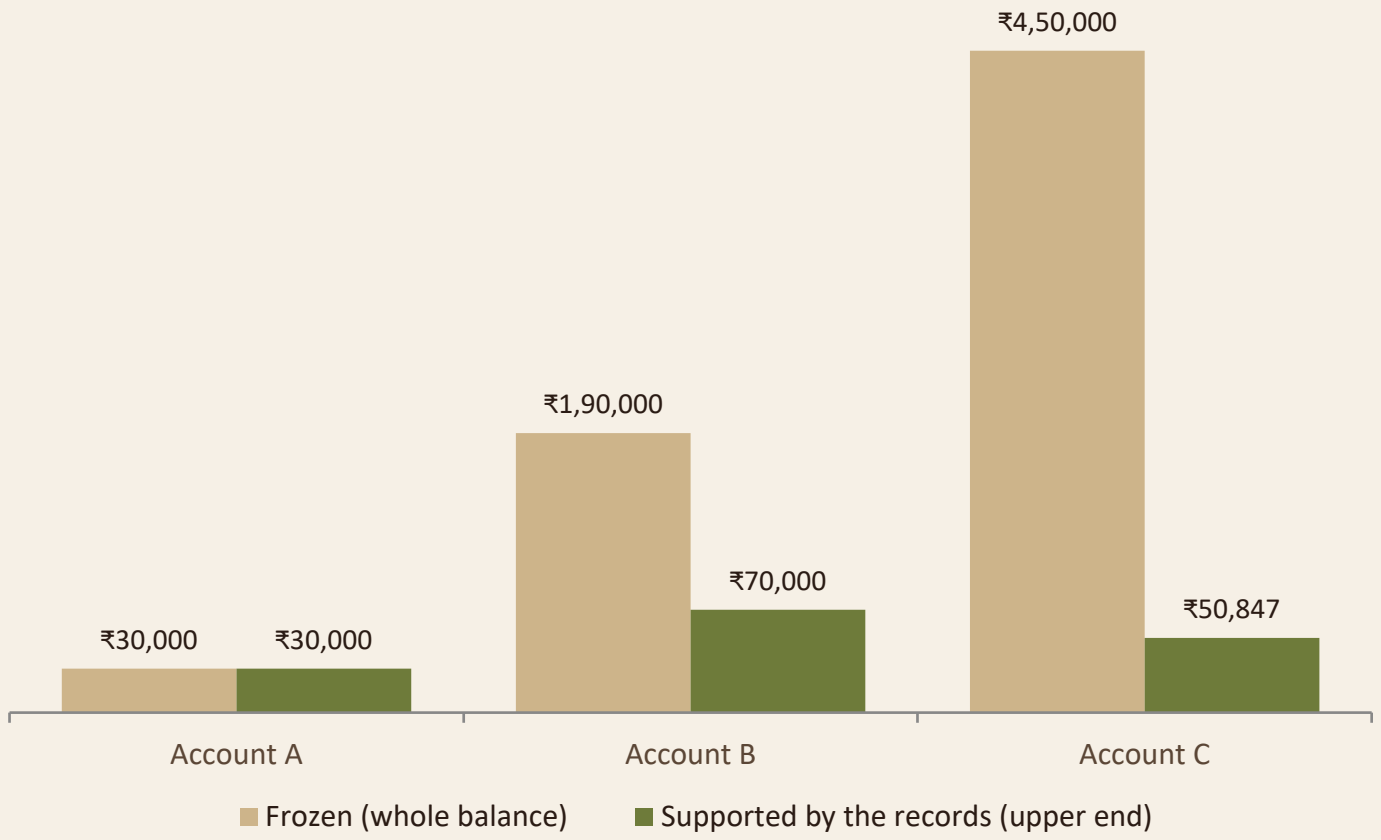
Holds outlive their purpose

Freezes continue after closure; holds above ₹50,000 have no timeline.

The rule already exists: hold the disputed sum, not the account (MHA procedure, Jan 2026). What is missing is the calculator. TraceHold is that calculator.

TraceHold sizes the hold the records can defend

₹1,00,000 is stolen and passes through three accounts. The police ask to freeze all three.

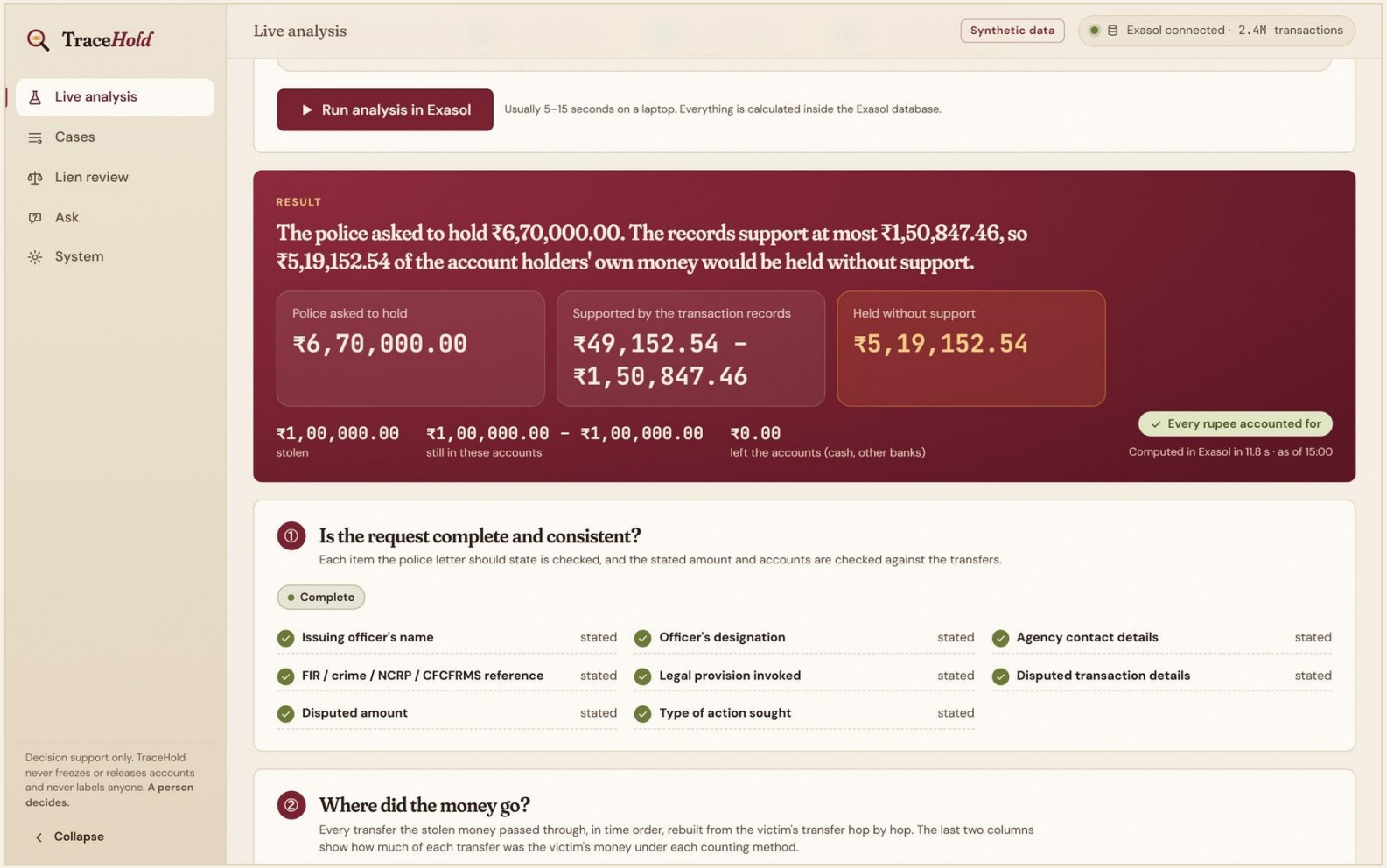


₹5,19,153

of the account holders' own money does not need to be restricted

- Every account gets a range under two disclosed methods (pro-rata and FIFO), a status, and the full calculation.
- A proportionate hold instruction is drafted for the officer.
- Every rupee of the loss is accounted for: a conservation check.

The verdict comes first: asked to hold vs supported by the records



1 **Pick a case or paste a police letter**

Five demo cases, your own values, or a free-text letter read with AI.

2 **Run it inside Exasol**

Around 10 seconds; every figure computed by SQL.

3 **Read the verdict, then the six steps**

Request check, money trail, sizing, comparison, hold draft, and the SQL that ran.

HOW IT WORKS

From a vague request to a figure you can defend

01

Check the request

Checklist from court directions; AI reads free-text letters, rules verify every field

02

Rebuild the trail

Hop by hop, through splits, merges, loops, reversals, duplicates and clock skew

03

Size every account

Pro-rata and FIFO ranges, traceable share, status; abstains when records are missing

04

Compare with the request

Within evidence, within range, exceeds evidence, or cannot determine

05

Hand over the evidence

Evidence packet, plain-language letter, proportionate hold, forwarded requests

A person decides at every step. TraceHold never freezes, releases or labels anyone, and no AI model ever produces a figure.

WHAT THE OFFICER GETS

Everything needed to size, review and explain a hold

Live analysis

Type in any case or paste a police letter; the verdict comes first: asked to hold vs supported by the records.

Every case, explained

2,400 cases on 2.4M transactions, each with the step-by-step maths. Re-trace any case live on the full ledger in about 10 s.

Lien review queue

Existing holds ranked for a person: over-held, closed but still frozen, past 90 days, no update.

Ask the results

Plain-English questions answered through Exasol's MCP server; the SQL and rows are always shown.

Proportionate hold draft

Per account, a hold limited to the supported range, ready for the officer to review and send.

Evidence for everyone

Evidence packet citing every transaction, and a neutral letter for the account holder.

Exasol is the engine, not just the storage

Set-based SQL, no row-by-row code. Window functions rebuild every running balance; the trail expands one layer per pass until nothing changes.

The whole analysis as one script. EXECUTE SCRIPT TRACEHOLD.RUN_ANALYSIS(): a Lua script generated from the same SQL; same checksum.

Governed questions. Exasol's open-source MCP server, as a read-only user that sees only results views.

Privileges as protection. The analysis user cannot change source data or history; Exasol refuses it.

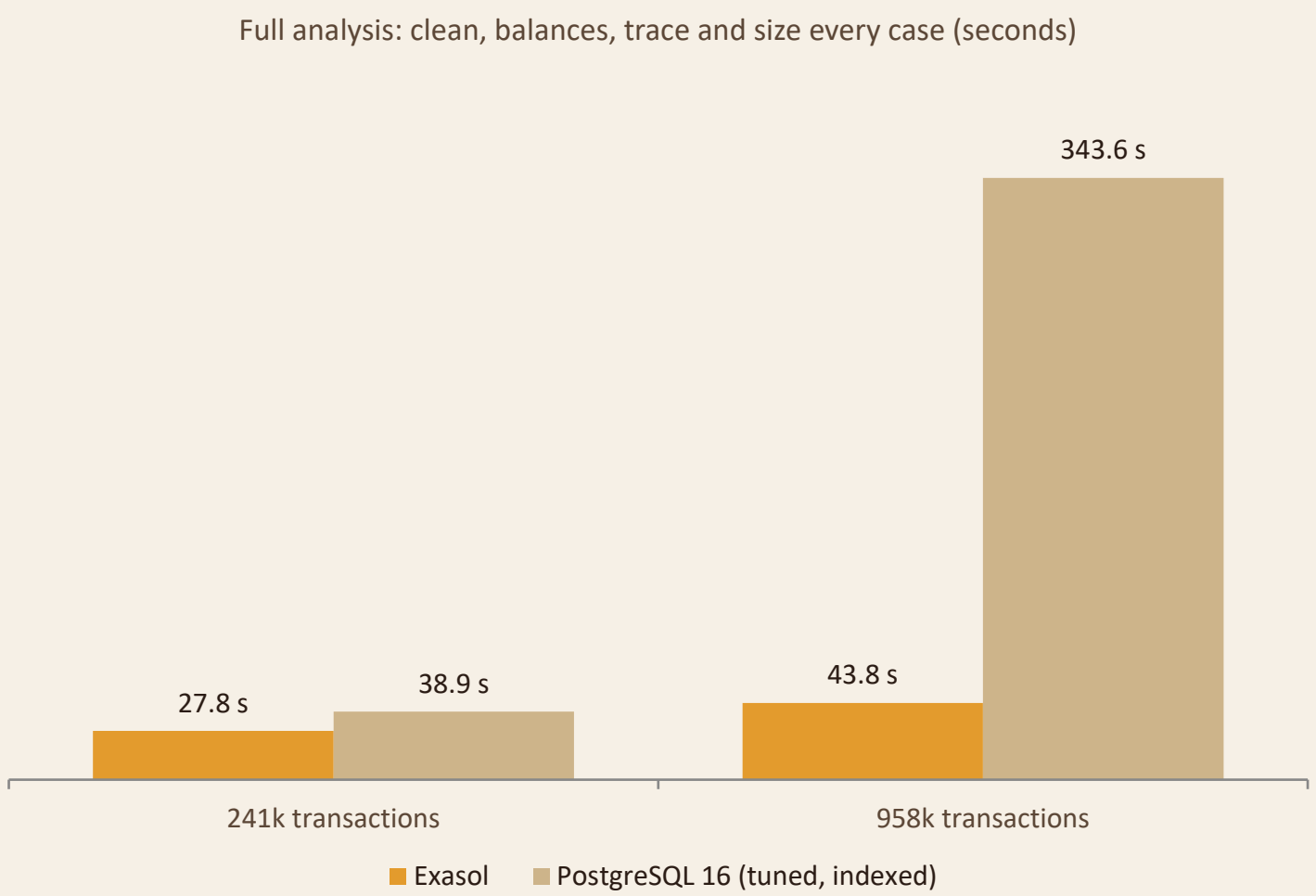
Exasol does the tuning. Every index is created by Exasol itself, none by hand; transactions compressed 2.3× in memory.

2.4M
transactions analysed in one run

16.7M
rows processed in one analysis run

≈ 4 min
for the complete analysis of 2,400 cases

Same SQL, same laptop: Exasol pulls ahead as data grows



1.4×
faster at 241k transactions

7.8×
faster at 958k transactions

With 4× more data, Exasol took 1.6× longer and PostgreSQL 8.8× longer. PostgreSQL is faster on tiny single queries; both answer those within seconds. Synthetic data; only these workloads were measured.

Protected data, and AI only at the edges

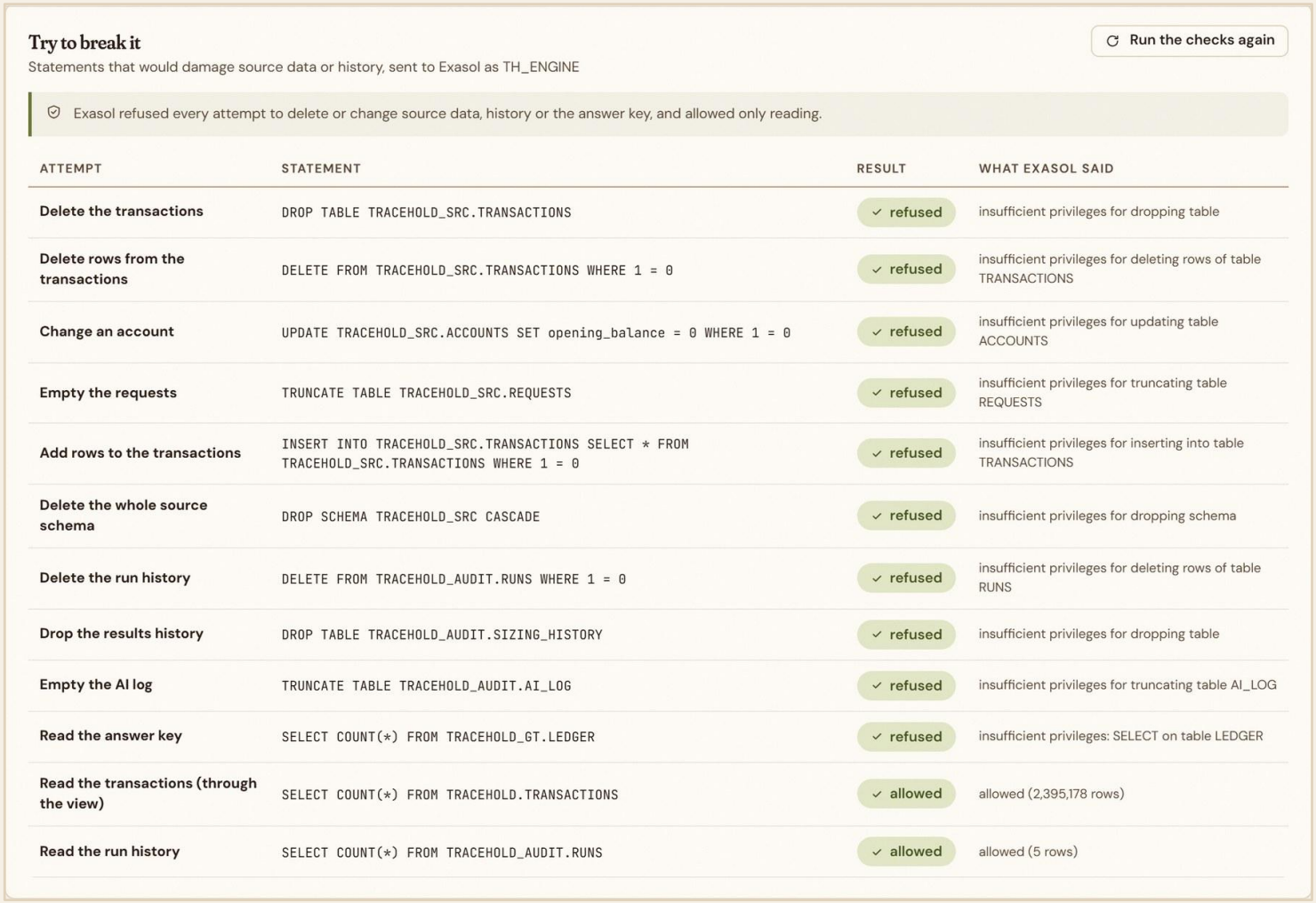
The records cannot be damaged

- Source data is read-only for the analysis: enforced by Exasol privileges, not by code alone
- History is append-only: every run's results, every live analysis, every AI call
- Each run fingerprints the source data before and after, and fails if anything changed
- Live proof in the app: 10 damaging statements sent, 10 refused by Exasol

AI never touches a figure

- AI reads free-text police letters; every value must quote the letter and pass rule checks
- Held-out letters: 98.8% precision, 99.2% recall, 0 values used without a quote
- Answers over results: every number must appear in the rows, or the wording is rejected
- With AI switched off, every figure is identical (tested)

Try to break it: Exasol refuses every damaging statement



Try to break it

Run the checks again

Statements that would damage source data or history, sent to Exasol as TH_ENGINE

Exasol refused every attempt to delete or change source data, history or the answer key, and allowed only reading.

ATTEMPT	STATEMENT	RESULT	WHAT EXASOL SAID
Delete the transactions	`DROP TABLE TRACEHOLD_SRC.TRANSACTIONS`	✓ refused	insufficient privileges for dropping table
Delete rows from the transactions	`DELETE FROM TRACEHOLD_SRC.TRANSACTIONS WHERE 1 = 0`	✓ refused	insufficient privileges for deleting rows of table TRANSACTIONS
Change an account	`UPDATE TRACEHOLD_SRC.ACCOUNTS SET opening_balance = 0 WHERE 1 = 0`	✓ refused	insufficient privileges for updating table ACCOUNTS
Empty the requests	`TRUNCATE TABLE TRACEHOLD_SRC.REQUESTS`	✓ refused	insufficient privileges for truncating table REQUESTS
Add rows to the transactions	`INSERT INTO TRACEHOLD_SRC.TRANSACTIONS SELECT * FROM TRACEHOLD_SRC.TRANSACTIONS WHERE 1 = 0`	✓ refused	insufficient privileges for inserting into table TRANSACTIONS
Delete the whole source schema	`DROP SCHEMA TRACEHOLD_SRC CASCADE`	✓ refused	insufficient privileges for dropping schema
Delete the run history	`DELETE FROM TRACEHOLD_AUDIT.RUNS WHERE 1 = 0`	✓ refused	insufficient privileges for deleting rows of table RUNS
Drop the results history	`DROP TABLE TRACEHOLD_AUDIT.SIZING_HISTORY`	✓ refused	insufficient privileges for dropping table
Empty the AI log	`TRUNCATE TABLE TRACEHOLD_AUDIT.AI_LOG`	✓ refused	insufficient privileges for truncating table AI_LOG
Read the answer key	`SELECT COUNT(*) FROM TRACEHOLD_GT.LEDGER`	✓ refused	insufficient privileges: SELECT on table LEDGER
Read the transactions (through the view)	`SELECT COUNT(*) FROM TRACEHOLD.TRANSACTIONS`	✓ allowed	allowed (2,395,178 rows)
Read the run history	`SELECT COUNT(*) FROM TRACEHOLD_AUDIT.RUNS`	✓ allowed	allowed (5 rows)

10 / 10

attempts to delete or change source data, history or the answer key refused

- Sent live from the app as the analysis user, on System → Runs & history.
- DROP, DELETE, UPDATE, TRUNCATE and INSERT on source tables, run history and the answer key.
- Only reading is allowed, and the reply comes from Exasol itself: "insufficient privileges".

RESULTS

Measured on 2,400 synthetic cases and 2.4M transactions

7,200 / 7,200

conservation checks pass: every rupee accounted for

99.9%

of account figures within one paisa of an independent ledger

100%

precision and recall on 946 planted request defects (14 types)

99.4% / 99.9%

precision / recall of money-trail reconstruction

3,588 of 3,589

over-broad requests flagged, with no false flags

Same checksum

on every re-run of the same data, either way it is run

The worked example from the problem statement is reproduced to the paisa. All results are on synthetic data with a separate answer key the analysis never reads; they are not evidence of real-world performance.

Others trace or detect. TraceHold sizes and reviews the hold

	CFCFRMS / NCRP	MuleHunter.AI	Fund-trace tools	TraceHold
Coordinate reporting across banks and police	✓	—	—	uses its output
Detect suspicious (mule) accounts	—	✓	—	not by design
Trace the money across accounts	—	—	✓	✓
Size the supportable hold per account	—	—	—	✓
Show uncertainty as a range under stated methods	—	—	—	✓
Check the request and flag over-broad holds	—	—	—	✓
Review existing holds by size and age	—	—	—	✓
Evidence packet citing every transaction	—	—	reports	✓

Based on public descriptions of existing systems; capabilities inside banks may differ. TraceHold does not claim to be first.

RESPONSIBLE USE

Careful by design, honest about its limits

1

A person decides

TraceHold never places, changes or releases a hold, and never decides guilt.

2

No labels, no scores

It reports facts for exceptional cases and never calls anyone a mule.

3

Conventions are estimates

Pro-rata and FIFO are disclosed methods, not legal findings about ownership.

4

Abstains when unsure

Missing records give 'Undeterminable', never a guessed number.

5

One bank's view

Money leaving the records gets a forwarded request, not a guess.

6

Synthetic data only

Patterns mirror the real problem in shape only; rules follow commentaries to be checked against primary texts.

Open source, reproducible, runs on a laptop

Database and analytics	Exasol Personal (nano runtime in Docker)
Analysis	Set-based SQL and an Exasol Lua script
Backend	Python, FastAPI, pyexasol
AI (optional)	Groq model; Exasol MCP server
Frontend	React, TypeScript, Vite
Quality	47 tests, 54 browser checks, WCAG AA checked

RUN IT

1. Generate synthetic data
2. Load it into Exasol (protected layout created)
3. Run the analysis inside Exasol
4. Start the API and open the website

github.com/lisamehta0791/TraceHold
MIT licence · README with the full run guide and screenshots

References

1. ExaThon India 2026 challenge page: exasol.com/events/exathon-india-2026
2. Deccan Herald: Bengaluru citizens lose over ₹1,200 crore to online fraud with low recovery rate
3. Ministry of Home Affairs, reply in the Rajya Sabha on CFCFRMS (Feb 2026)
4. Pink City Post: Rajasthan High Court directions on lien versus blanket freeze (Aug 2026)
5. LiveLaw: Problem with CFCFRMS, reading MHA's new account-freeze SOP (2 Jan 2026 procedure)
6. MediaNama: RTI reply on MuleHunter.AI adoption by 23 banks (Dec 2025)
7. Voice&Data: iAcuity launches Fund Trail (Mar 2023)
8. Exasol: MCP server, Exasol Personal, developer documentation

Court and procedure references are context, not legal advice; the Rajasthan order binds Rajasthan only. All project data is synthetic.

THANK YOU

Trace*Hold*

Hold the victim's money sooner. Leave everyone else's alone.



Scan for the code

github.com/lisamehta0791/TraceHold

Lisa Mehta · github.com/lisamehta0791/TraceHold · Built on Exasol Personal